Do Tabular Foundation Models Pay Off for Churn? A Profit-, Calibration- and Tuning-Aware Re-evaluation

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Abstract

Tabular foundation models (TFMs) predict from a labelled context in a single forward pass. A recent benchmark reported that they decisively outperform tree ensembles for customer churn, but it compared them with untuned gradient-boosted trees (GBDTs) and scored only ranking metrics, while a retention campaign is judged in money. We re-evaluate TabPFN-2.5, TabPFN-3.5, TabPFN-3.5-Fast and TabICL v2 against Optuna-tuned GBDTs on five public telecom churn datasets over 10 seeded splits. We separate three questions: whether a model ranks churners well, whether that ranking earns more expected maximum profit (EMP), and whether its probabilities can drive a threshold policy. With a split-aware test and Holm correction over 20 pre-declared comparisons, TabPFN-3.5 earns significantly more EMP than the best tuned GBDT on Cell2Cell (1.9%) and Iranian (7.6%). On Orange it wins in all 10 splits (14.3%), but the gain does not survive correction. TabICL v2, the earlier benchmark's leader, is significant only on Iranian and underpredicts Orange churn by 2.1 points on a 7.3% base rate. On the datasets and splits where we tested them, a fivefold larger tuning budget and a duplicate-aware split leave these conclusions unchanged. Retuning the baseline cuts the earlier headline gap from 6.1 to 4.0 PR-AUC points without closing it. The licence of TabPFN-3.5's weights, not scoring cost, is what stands between its gains and production use.

1 Introduction

Churn models exist to choose whom a retention campaign contacts. A model is useful to an operator to the extent that the customers it ranks highest are the ones worth an incentive, so the natural yardstick is campaign profit rather than classification accuracy. Verbraken et al. made this precise with the expected maximum profit (EMP) criterion [36], and later work showed that selecting churn models by AUC instead of profit leaves money on the table [16, 20, 35].

Tabular foundation models change what a churn model is. TabPFN [14, 24, 26] and TabICL [28, 29] are pretrained on synthetic tables and predict a new customer by conditioning on the labelled training set as context, with no fitting and no hyperparameter search. On general benchmarks they now lead or match tuned GBDTs [12, 26, 29]. For churn specifically, Seyedzadeh and Karimi [30] benchmarked six TFMs on nine Kaggle datasets. They concluded that TFMs "collectively and decisively outperform" classical, deep and tree-ensemble models, with TabICL v2 ahead of XGBoost by up to 9.23 PR-AUC points.

That study was the first to bring TFMs to churn, and it leaves three questions open. Its XGBoost and random forest used library-default hyperparameters [30, Section 3.2], so part of the reported gap may be a tuning gap. Its metrics were F1, ROC-AUC, PR-AUC and log loss; its financial estimate scaled the PR-AUC gap by a retention rate rather than evaluating a campaign. Finally, its own discussion reads TabICL v2's lead as "operational ... rather than as a statistically decisive metric lead" [30, Section 4.2], which leaves the size of the lead unsettled.

We argue that the practical value of a TFM for churn depends on the model generation, the dataset, the decision policy and the deployment terms, not on membership of the model class. We test this with tuned baselines and an explicit campaign model. Four TFMs (TabPFN-2.5, TabPFN-3.5, TabPFN-3.5-Fast, TabICL v2) face Optuna-tuned LightGBM, XGBoost and CatBoost, an explainable boosting machine and logistic regression on five public telecom datasets, three of them from real operators, over 10 seeded splits. Our inference treats the split as a source of variation and corrects for multiple comparisons. The contributions are:

1. **A tuned, profit-aware comparison.** TabPFN-3.5 earns significantly more EMP than the best tuned GBDT on Cell2Cell and Iranian after Holm correction, wins in every split on Orange without surviving correction, and is never worse beyond noise. TabICL v2 and TabPFN-2.5 are significant only on Iranian. On the two fictional datasets, Telco and Maven, no difference survives correction (Section 4.1).

2. **Ranking, profit and probabilities kept apart.** Profit metrics rerank models only when they are tied (Section 4.2). Calibration matters for one model on one dataset: raw TabICL v2 loses 15.7% of attainable campaign profit on Orange at a fixed threshold (Section 4.3).

3. **Robustness to the obvious objections.** A balanced context hurts rather than helps when the training set fits (Section 4.4). A fivefold larger tuning budget (on the three real-operator datasets) and a split that keeps duplicates without leaking them (on the three datasets that had any) change no conclusion within the five splits each check covers (Section 4.5).

4. **A bounded replication.** On the earlier benchmark's headline dataset, tuning removes about a third of TabICL v2's PR-AUC lead, and the rest survives (Section 4.6).

5. **Deployment terms.** Cached scoring is cheap; the TabPFN weights' licence, which excludes production use, is the binding constraint (Section 4.7).

The code, every per-split prediction, and the generator and audit for every number in this paper are released (Section 7).

2 Related Work

Profit-aware churn evaluation. Verbraken et al. defined EMP as the expected best campaign profit when the probability that a contacted churner accepts the offer is uncertain and Beta distributed [36]. Verbeke et al. used profit to compare churn classifiers in telecom [35], and Höppner et al. optimised EMP directly when growing decision trees [16]. Manzoor et al. proposed e-Profits, which replaces EMP's population-level customer value with a per-customer value derived from Kaplan–Meier retention, and found that it reranks six classical models on IBM Telco and Maven [20]. None of these studies includes a foundation model.

Tabular foundation models. TabPFN v2 [14] established prior-fitted transformers for tables up to roughly 10k rows; TabPFN-2.5 extended this to 50k rows [24], and TabPFN-3 and 3.5 to a million rows with a cached context for fast scoring [25, 26]. TabICL separates column and row attention to scale in-context learning [28]; TabICL v2 reports beating a tuned and ensembled RealTabPFN-2.5 on TabArena and TALENT without tuning [29]. On TabArena, GBDTs remain strong on larger datasets while TFMs excel on small ones [12], and TabPrep shows that feature engineering, absent from most benchmarks, changes these comparisons [34]. TabPFN's predictive probabilities are reported to be well calibrated out of the box [31], while ensembling TFMs can damage calibration [32].

Table 1: Closest prior studies. Tuned: baselines hyperparameter-tuned. Calibration: analysed beyond reporting log loss. Tests: uncertainty or significance reported for differences between models.

Study	Domain	TFMs	Tuned	Profit metric	Calibration	Tests
Manzoor et al. [20]	churn	no	yes	e-Profits, EMP	no	yes
Baesens et al. [10]	credit risk	yes	yes	no	yes	yes
Seyedzadeh & Karimi [30]	churn	yes	no	no	log loss only	no
This work	churn	yes	yes	EMP, e-Profits, policy	yes	yes

Table 2: Datasets after preprocessing. Dup.: exact duplicate feature rows removed. e-Prof.: tenure and monthly revenue available for e-Profits.

Dataset	Source	Rows	Feat.	Churn %	Dup.	e-Prof.	Role
Telco [7]	IBM sample data (fictional)	7,003	19	26.4	40	yes	main
Maven [3]	Maven Analytics (fictional)	6,589	34	28.4	0	yes	main
Cell2Cell [1]	US wireless carrier	51,047	56	28.8	0	yes	main
Iranian [8]	Iranian telecom operator	2,836	15	15.6	314	no	main
Orange [13]	Orange, KDD Cup 2009	50,000	202	7.3	0	no	main
E-Commerce [2]	Kaggle, origin unstated	5,074	18	16.6	556	no	replication

TFMs in business prediction. The closest analogue to our study is the credit-risk review of Baesens et al. [10]. Across 14 probability-of-default datasets TabPFN-3 ranks first, but its margin over the strongest GBDT is narrow and "driven by the latest TabPFN and TabICL generations rather than by the model class as a whole". TabPFN-3 also underpredicts the mean default rate. Tanna et al. found that, under heavy imbalance, building a class-balanced context adds 3 to 4 AUC points over uniform sampling for subsampled contexts of 1k to 50k rows [33]; V4FinBench reports a similar benefit of imbalance-aware context construction for bankruptcy prediction [19]. For churn, the only TFM benchmark is [30]. Table 1 sets the closest studies against ours.

3 Benchmark Design

3.1 Datasets

Table 2 lists the six datasets. Five form the main benchmark. Cell2Cell, Iranian and Orange come from real operators: Cell2Cell from a US wireless carrier [1], Iranian from an Iranian operator through the UCI repository [8], and Orange from the KDD Cup 2009 churn task [13], which we load from OpenML (dataset 1112). IBM Telco [7] and Maven [3] are fictional teaching datasets; we keep them because prior churn studies use them, including both [20] and [30], and because a gap between real and fictional data would itself be informative. E-Commerce [2], the dataset on which [30] reported its largest gain, serves only as a replication check (Section 4.6) and is excluded from all cross-dataset summaries.

All models share one preprocessing pipeline. Categorical columns are ordinal-encoded with missing values left as missing, and columns that are more than 99% missing or constant are dropped. For Maven we drop customers whose status is "Joined", whose outcome is not yet observed, and the two columns that record the churn reason after the fact. We remove exact duplicate feature rows, keeping the first, because an in-context learner can otherwise find a test customer's twin in its own context. This removed 40 rows (0.6%) from Telco, 314 (10.0%) from Iranian and 556 (9.9%) from E-Commerce; the other datasets had none. The removal changed churn rates by at most 0.3 points, and Section 4.5 repeats the analysis with duplicates kept.

3.2 Models

Foundation models. TabPFN-2.5, TabPFN-3.5 and TabPFN-3.5-Fast run through the `tabpfn` package (v9.2.0) with default settings and the training-size guardrail disabled; TabICL v2 runs through `tabicl` (v2.2.0, checkpoint of 2026-02-12). Each receives the full training split as its context, at the natural churn rate. None is tuned, which is how these models are meant to be used.

Tuned baselines. LightGBM [18] and XGBoost [11] are tuned with 40 trials of Optuna's TPE sampler [9], and CatBoost [27] with 25, each under a 20-minute cap. Trials are scored by validation AUC on an inner 80/20 split of the training data with early stopping; the best configuration is then refit on the whole training split with the selected number of trees. Search spaces are listed in Appendix A. We also report an explainable boosting machine (EBM) [22] and logistic regression with one-hot encoding and median imputation, both at defaults.

3.3 Protocol and inference

For each dataset and each of 10 seeds we draw a stratified 60/20/20 split into training, calibration and test sets. All models see identical splits, which the released audit verifies. Models are fitted (or given context) on the training split; the calibration split is used only to fit post-hoc recalibration; every reported metric is computed on the test split. GBDTs ran on Kaggle CPU sessions and TFMs on a single NVIDIA T4 (16 GB).

The estimand is the difference in a test-set metric between a TFM and a baseline, averaged over random splits. The *primary family* is fixed in advance: EMP, each of the four TFMs against the best tuned GBDT on each of the five main datasets, 20 comparisons in all. The reference GBDT is chosen per dataset and metric as the tuned GBDT with the best mean score on the calibration split, so the test set plays no part in choosing it. Two sensitivity analyses address the remaining selection: choosing the reference by its test mean instead, and an intersection–union test that requires a TFM to beat each of the three tuned GBDTs separately, taking the largest of the three p-values, again Holm-corrected over the 20 comparisons. Because random splits share training and test customers, the seed-level differences are correlated, and a plain t-test over seeds understates their variance. We therefore use the corrected resampled t-test of Nadeau and Bengio [21], which inflates the variance of the mean difference by $1/J + n_{\text{test}}/n_{\text{train}}$ for J splits, and apply Holm's correction [15] across the primary family. We call a primary comparison significant when its Holm-adjusted p is below 0.05, and report the corrected test's 95% confidence interval, which is unadjusted for multiplicity. A two-level bootstrap that resamples seeds and then test customers is computed as a descriptive check; it does not model the overlap between splits, so it is narrower than the corrected test, and no claim rests on it. Everything outside the primary family, including PR-AUC, the ablations and the replication, is exploratory and reported with unadjusted p.

3.4 Metrics: three questions

We keep three questions apart, because a model can answer one well and another badly.

Does the model rank churners well? ROC-AUC, PR-AUC and lift in the top decile.

Does the ranking earn more campaign profit? For a campaign that contacts the fraction of customers with the highest scores, a contacted churner accepts the offer with probability γ and is then worth $\text{CLV} - \delta - \phi$; a churner who declines costs the contact cost ϕ; a contacted non-churner takes the incentive and costs $\delta + \phi$. EMP is the expected profit per customer at the profit-maximising cutoff, averaged over $\gamma \sim \text{Beta}(6, 14)$ [36], with the standard $\text{CLV} = 200$, $\delta = 10$ and $\phi = 1$. Because EMP picks the best cutoff for each model, it depends only on the ranking: a high EMP does not mean the probabilities can be used directly. We also report e-Profits [20], which values each customer as $\text{CLV}_i = R_i M/(1 - \min(r_i, 0.995))$ from

monthly revenue R_i, margin $M = 0.3$ and one-month retention $r_i = S(T_i + 1)/S(T_i)$ from a Kaplan–Meier curve S [17] fitted on the training split. The offer costs $0.1\,\mathrm{CLV}_i$ and the contact $\max(5, 0.3 \cdot \text{offer})$. We report it when contacting the top 20% and, as in [20], when contacting customers with predicted probability at least 0.5; the latter mixes ranking with calibration.

Can the probabilities drive a threshold policy? An operator who acts on probabilities contacts customer i when its expected profit under the EMP cost model is positive at the mean acceptance rate $\bar{\gamma} = 0.3$. We report the policy's regret: the profit lost relative to the best cutoff on the same ranking, as a share of that best profit, as the median over a 3×3 grid of CLV $\in \{100, 200, 500\}$ and $\delta \in \{5, 10, 20\}$. Calibration is measured by expected calibration error (ECE, 15 equal-width bins), Brier score, and calibration-in-the-large: the observed churn rate minus the mean predicted probability, in percentage points, positive when churn is underpredicted. Each model is evaluated raw and after isotonic regression [37] or Platt scaling [23], both fitted on the calibration split.

4 Results

4.1 Do foundation models earn more profit than tuned GBDTs?

TabPFN-3.5 earns significantly more EMP than the best tuned GBDT on two of five datasets (Table 3, Figure 1). On Cell2Cell the gain is 0.175 per customer (1.9%; Holm p <0.001), and on Iranian 0.597 (7.6%; Holm $p = 0.003$). On Iranian every TFM is significantly ahead, which reflects how far the boosted models fall behind there rather than anything specific to TabPFN. In total, 6 of the 20 primary comparisons are significant after correction, and none favours the GBDT. The selection of the reference does not drive this: choosing it on test means instead gives 6 significant comparisons, and the stricter intersection–union test against all three tuned GBDTs gives 3 (TabPFN-3.5 on Cell2Cell; TabPFN-3.5 on Iranian; TabPFN-3.5-Fast on Cell2Cell).

Orange is the case the correction decides. TabPFN-3.5 beats the best tuned GBDT in all 10 splits, by 0.082 per customer, which is 14.3% of an EMP that is small because only 7.3% of Orange customers churn. Its corrected p is 0.011 before adjustment and 0.155 after. We read this as a consistent gain whose size is not established; the relative figure in particular should not be carried to other operators.

On the two fictional datasets no difference survives correction. TabPFN-3.5 is ahead by 1.0% on Telco and 1.0% on Maven, with Holm-adjusted p of 1. TabICL v2 is significant only on Iranian, and on Orange its mean EMP is below the GBDT's (-4.5%, not significant). Version matters more than family: TabPFN-2.5 and TabPFN-3.5 share a lineage, yet only the latter separates from the GBDTs on Cell2Cell, which matches the pattern reported for credit risk [10]. PR-AUC differences, treated as exploratory, point the same way (Appendix, Table 12).

4.2 Do profit metrics change which model wins?

They change it when models are tied and leave it alone otherwise (Table 4). On Cell2Cell and Orange, where TabPFN-3.5 leads clearly, the EMP winner differs from the AUC winner in only 0 and 1 of 10 splits. On Telco and Maven, where no difference is significant, the EMP winner differs from the AUC winner in 7 and 7 of 10 splits. Iranian sits in between (6 of 10): the four TFMs are tied with each other there, though all lead the GBDTs.

e-Profits at the 0.5 threshold disagrees with AUC the most ($\tau = 0.04$ on Maven). This follows from its construction rather than from a better ranking: at a fixed threshold the metric also rewards a model whose probabilities happen to fall on the right side of 0.5. The reranking reported for e-Profits on Telco and Maven [20] is therefore reproduced here, among models whose rankings are statistically indistinguishable.

Table 3: Primary family: EMP of each TFM minus the reference GBDT (best tuned GBDT on the calibration split), per customer (CLV units) and relative to the GBDT. CI: 95% interval of the corrected resampled t-test, unadjusted for multiplicity. Seeds: splits in which the TFM is ahead. p: corrected test; Holm: adjusted over the 20 comparisons. All: Holm-adjusted intersection–union p against each of the three tuned GBDTs. Bold: below 0.05. Significance is decided only by the Holm-adjusted columns; an unadjusted CI that excludes zero is not a claim.

Dataset	TFM	ΔEMP	%	95% CI (corrected)	Seeds	p	Holm	All
Telco	TabICL v2	+0.066	+0.6	[−0.024, +0.156]	9/10	0.133	1.000	1.000
	TabPFN-2.5	+0.060	+0.6	[−0.048, +0.168]	8/10	0.240	1.000	1.000
	TabPFN-3.5	+0.104	+1.0	[−0.014, +0.222]	9/10	0.078	1.000	1.000
	TabPFN-3.5-Fast	+0.090	+0.9	[−0.020, +0.200]	9/10	0.097	1.000	1.000
Maven	TabICL v2	+0.042	+0.3	[−0.129, +0.213]	6/10	0.589	1.000	1.000
	TabPFN-2.5	+0.090	+0.7	[−0.083, +0.264]	8/10	0.270	1.000	1.000
	TabPFN-3.5	+0.135	+1.0	[−0.041, +0.312]	8/10	0.116	1.000	1.000
	TabPFN-3.5-Fast	+0.095	+0.7	[−0.070, +0.260]	8/10	0.225	1.000	1.000
Cell2Cell	TabICL v2	+0.042	+0.5	[−0.007, +0.091]	9/10	0.086	1.000	1.000
	TabPFN-2.5	+0.032	+0.4	[−0.012, +0.077]	8/10	0.136	1.000	1.000
	TabPFN-3.5	+0.175	+1.9	[+0.133, +0.216]	10/10	<0.001	**<0.001**	**<0.001**
	TabPFN-3.5-Fast	+0.104	+1.1	[+0.048, +0.161]	10/10	0.002	**0.036**	**0.046**
Iranian	TabICL v2	+0.584	+7.4	[+0.333, +0.835]	10/10	<0.001	**0.009**	0.051
	TabPFN-2.5	+0.543	+6.9	[+0.302, +0.785]	10/10	<0.001	**0.011**	0.093
	TabPFN-3.5	+0.597	+7.6	[+0.382, +0.813]	10/10	<0.001	**0.003**	**0.043**
	TabPFN-3.5-Fast	+0.561	+7.1	[+0.317, +0.806]	10/10	<0.001	**0.010**	0.082
Orange	TabICL v2	−0.026	−4.5	[−0.101, +0.049]	4/10	0.457	1.000	1.000
	TabPFN-2.5	−0.017	−3.0	[−0.070, +0.035]	2/10	0.474	1.000	1.000
	TabPFN-3.5	+0.082	+14.3	[+0.024, +0.140]	10/10	0.011	0.155	0.155
	TabPFN-3.5-Fast	+0.045	+7.8	[−0.011, +0.101]	8/10	0.102	1.000	1.000

4.3 Can the probabilities drive a threshold policy?

For most models, yes. Outside Orange, the fixed-threshold policy loses at most 3.1% of attainable profit without recalibration, and isotonic or Platt scaling changes that loss by at most 0.9 points (Table 5).

Orange is the exception, and one model drives it. Raw TabICL v2 predicts 2.1 percentage points too little churn on a base rate of 7.3% (Figure 2), and the policy built on those probabilities loses 15.7% of attainable profit. Isotonic regression cuts the loss to 7.5% and Platt scaling to 6.3%. TabICL v2 also underpredicts on Cell2Cell (by 1.8 points), as does TabPFN-2.5 (2.0). TabPFN-3.5's calibration-in-the-large stays within ±0.2 points on every dataset, so it is the only TFM here whose raw probabilities can drive a threshold policy. The tuned GBDTs are close to zero as well, consistent with their fitting to the observed base rate.

4.4 Does a balanced context help?

Under heavy imbalance, a balanced context has been reported to add 3 to 4 AUC points for TFMs [33]. If that held here, our full-context runs would understate the TFMs on Orange. For TabPFN-3.5 and TabICL v2 on Orange and Cell2Cell we kept every churner in the training split and sampled one or two non-churners per churner, on the first five splits.

Balancing hurt (Table 6). TabPFN-3.5's EMP gain on Orange fell from 14.3% with the full context to 8.2% at 1:2 and 4.4% at 1:1 (unadjusted p 0.201 and 0.541). TabICL v2 gained nothing on either dataset. Balancing also inflated raw predicted churn, by 35.3 points on Orange at 1:1, which isotonic recalibration removes. The difference from [33] is the regime: there, contexts were subsampled to at most 50k rows from larger data, so balancing chose which customers to keep. Here the whole training split fits, and undersampling only discards non-churners.

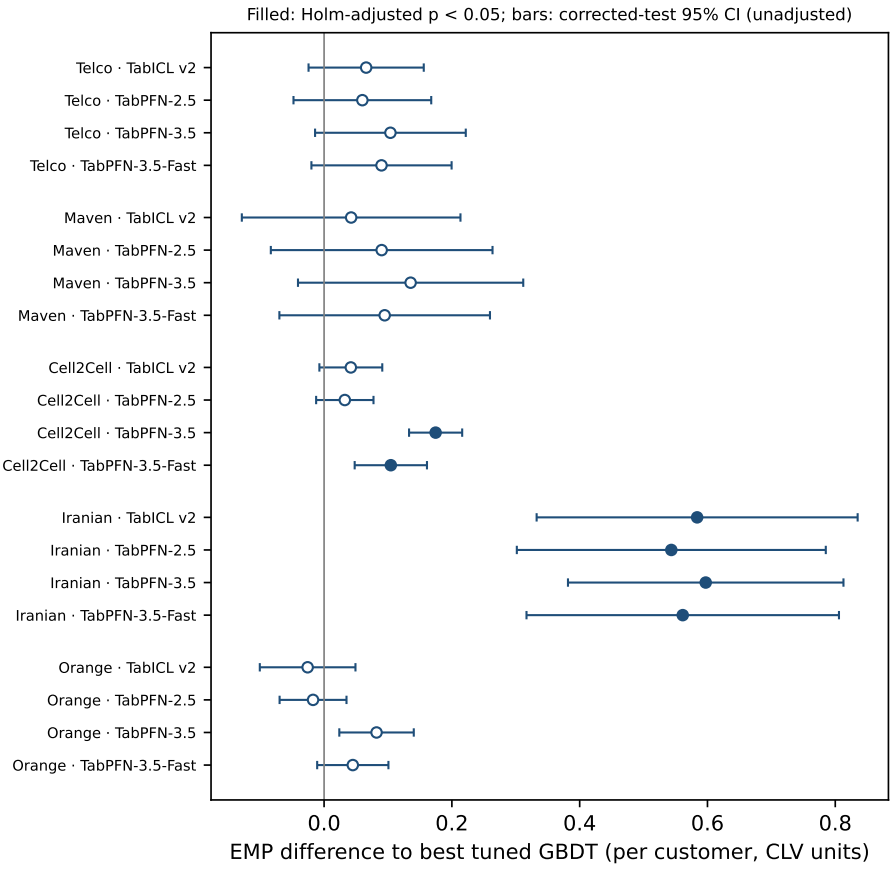


Figure 1: EMP difference between each TFM and the reference GBDT, per customer, with corrected-test 95% intervals (unadjusted) over 10 splits. Filled markers: significant after Holm correction across the 20 comparisons.

4.5 Robustness: tuning budget and duplicates

Tuning budget. Our baselines use 25 to 40 trials, fewer than benchmark practice [12]. We reran LightGBM and CatBoost with 200 trials each on Cell2Cell, Iranian and Orange for the first five splits. The larger budget changed the best GBDT's mean EMP by 0.0%, −0.8% and +0.1% respectively, and the TFM gains were unchanged (Table 7a). On Cell2Cell the 200-trial LightGBM becomes the reference and TabPFN-3.5 still gains 1.9% in each of the five splits.

Duplicates. Removing duplicates could discard legitimate customers. A duplicate group is a set of rows with identical features; removal keeps one row per group, discarding the other copies. In E-Commerce every group's copies share the same churn label. In Iranian, 14 groups (64 rows in total) and in Telco 18 groups (42 rows) contain copies with different labels, so keeping the first copy assigns them one label; removal changed churn rates by at most 0.3 points. We repeated the analysis keeping every row but assigning all copies of a feature vector to the same side of each split, so that no twin crosses from context to test. The gains moved by at most 1.6 points on all three datasets (Table 7b).

4.6 Does the earlier headline survive tuning?

We reran our protocol on E-Commerce, where [30] reported TabICL v2 9.23 PR-AUC points ahead of XGBoost, adding a library-default XGBoost to stand in for their baseline. This is a replication under a different protocol, not a reproduction (Table 8). TabICL v2 leads default XGBoost by 6.1 points and the best tuned GBDT, CatBoost, by 4.0 points, ahead in all 5

Table 4: Agreement between the AUC ranking of all nine models and profit-based rankings: mean Kendall τ over splits, and splits in which the top model changes. e-Profits needs tenure and revenue.

	EMP		e-Profits, top 20%		e-Profits, $p \geq 0.5$	
Dataset	τ	top-1 changed	τ	top-1 changed	τ	top-1 changed
Telco	0.52	7/10	0.23	10/10	0.28	10/10
Maven	0.67	7/10	0.47	5/10	0.04	7/10
Cell2Cell	0.88	0/10	0.42	1/10	0.41	1/10
Iranian	0.79	6/10	–	–	–	–
Orange	0.83	1/10	–	–	–	–

Table 5: Calibration. CITL: observed minus mean predicted churn rate (points; positive = underprediction). Regret: share of best-cutoff profit lost by the fixed-threshold policy (median over the cost grid), raw / isotonic / Platt. The first row per dataset is the best tuned GBDT.

			ECE		Policy regret (%)		
Dataset	Model	CITL	raw	iso.	raw	iso.	Platt
Telco	LightGBM	−0.1	0.051	0.029	1.8	1.3	1.4
	TabICL v2	+0.7	0.030	0.028	1.5	1.5	1.1
	TabPFN-2.5	+0.6	0.028	0.031	1.4	1.6	1.4
	TabPFN-3.5	0.0	0.031	0.029	1.3	1.7	1.2
Maven	XGBoost	+0.6	0.027	0.028	0.9	1.2	0.9
	TabICL v2	+0.8	0.026	0.024	1.1	1.1	0.9
	TabPFN-2.5	+0.7	0.022	0.026	0.9	1.0	1.1
	TabPFN-3.5	+0.2	0.020	0.023	0.7	1.0	0.9
Cell2Cell	LightGBM	+0.1	0.011	0.011	0.5	0.5	0.4
	TabICL v2	+1.8	0.019	0.012	0.6	0.5	0.5
	TabPFN-2.5	+2.0	0.021	0.012	0.7	0.4	0.5
	TabPFN-3.5	−0.2	0.012	0.014	0.5	0.5	0.6
Iranian	LightGBM	+0.8	0.033	0.024	2.3	2.6	2.1
	TabICL v2	−0.1	0.014	0.011	0.8	1.0	1.0
	TabPFN-2.5	0.0	0.016	0.011	1.0	0.9	0.8
	TabPFN-3.5	−0.1	0.013	0.012	0.9	0.9	0.9
Orange	LightGBM	0.0	0.008	0.005	8.2	7.1	5.8
	TabICL v2	+2.1	0.023	0.004	15.7	7.5	6.3
	TabPFN-2.5	+0.2	0.008	0.006	7.7	6.1	6.0
	TabPFN-3.5	0.0	0.005	0.005	5.5	5.7	4.9

splits (unadjusted p 0.023), with an EMP gain of 4.8% (Table 9). Tuning raised the GBDT's PR-AUC from 0.906 to 0.927 and so removed about a third of the gap. TabPFN-3.5 leads by more (5.8 points over tuned CatBoost; EMP gain 8.3%). On this dataset, then, TFMs win clearly. The origin of E-Commerce is not documented, so it cannot tell us whether the effect is specific to real or synthetic data.

4.7 Scoring cost and licensing

An in-context learner re-reads its context for every scoring call unless the context is cached, so its cost depends on how scoring is run. We timed scoring of 1k, 10k and 50k customers resampled from the test split, after a warm-up call, on one T4 (Table 10). With the context cached, TabPFN-3.5 scores 10k Orange customers in 45 s against a 30,000-row context, and TabICL v2 with its key-value cache in 17 s, against 0.8 s for a 1,000-tree LightGBM on CPU. A monthly scoring run of a million customers would therefore take about 1.25 T4-hours with cached TabPFN-3.5. This conclusion is conditional on batch scoring, a cached context and this hardware; real-time serving with tight latency budgets is a different question we did not

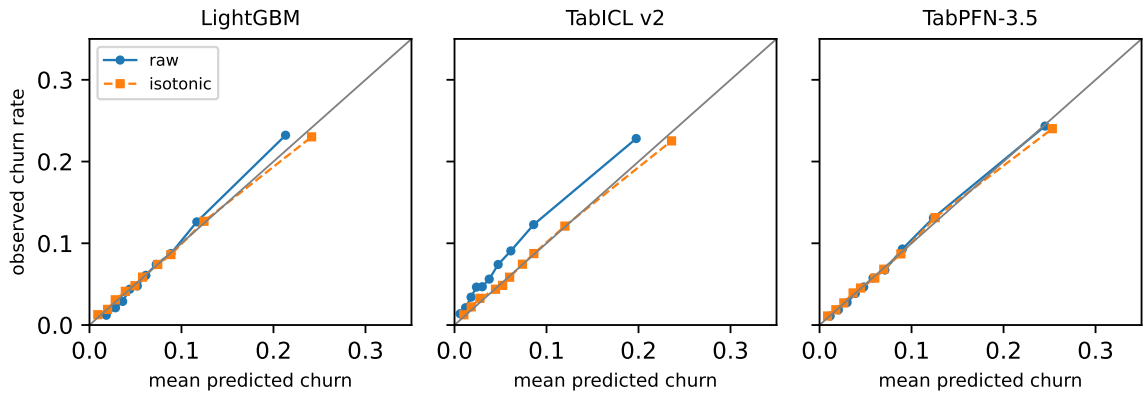


Figure 2: Reliability on Orange, test customers pooled over 10 splits, ten quantile bins. Raw TabICL v2 sits above the diagonal (churn underpredicted); isotonic recalibration on the calibration split removes the gap.

Table 6: Context construction (exploratory). Full: the whole training split at its natural churn rate (10 splits). 1:2 and 1:1: all churners plus two or one sampled non-churners per churner (five splits). ΔEMP relative to the best tuned GBDT, splits ahead, and unadjusted corrected p. CITL in points, raw.

Dataset	Model	Context	AUC	PR-AUC	EMP	ΔEMP	Seeds	p	CITL
Orange	TabPFN-3.5	full (natural rate)	0.748	0.238	0.658	+14.3%	10/10	0.011	0.0
Orange	TabPFN-3.5	1:2	0.739	0.219	0.611	+8.2%	5/5	0.201	−21.6
Orange	TabPFN-3.5	1:1	0.736	0.208	0.589	+4.4%	3/5	0.541	−35.3
Orange	TabICL v2	full (natural rate)	0.727	0.210	0.550	−4.5%	4/10	0.457	+2.1
Orange	TabICL v2	1:2	0.730	0.209	0.557	−1.2%	3/5	0.875	−19.9
Orange	TabICL v2	1:1	0.727	0.196	0.525	−6.8%	0/5	0.309	−35.1
Cell2Cell	TabPFN-3.5	full (natural rate)	0.702	0.495	9.350	+1.9%	10/10	<0.001	−0.2
Cell2Cell	TabPFN-3.5	1:2	0.704	0.498	9.357	+1.9%	5/5	0.006	−4.4
Cell2Cell	TabPFN-3.5	1:1	0.700	0.491	9.324	+1.5%	5/5	0.021	−19.1
Cell2Cell	TabICL v2	full (natural rate)	0.687	0.472	9.218	+0.5%	9/10	0.086	+1.8
Cell2Cell	TabICL v2	1:2	0.689	0.474	9.226	+0.4%	4/5	0.301	−2.7
Cell2Cell	TabICL v2	1:1	0.686	0.468	9.197	+0.1%	3/5	0.750	−19.1

measure. Without caching, small batches are very expensive (1,686 s per 10k for TabPFN-3.5 at 1k rows on Orange), and TabPFN-2.5 was impractical at 30k-row contexts.

The binding constraint is the licence. The TabPFN-3.5 and TabPFN-2.5 weights are released under non-commercial licences that forbid using the model, its derivatives "or any data produced by" the model "for any commercial or production purposes" [5, 6]; the licences define outputs to include predictions, scores and probabilities. The TabPFN-3.5 report states the practical consequence: outputs may not be used "as inputs to internal commercial decision-making" without a commercial licence [26, Section 4]. The TabICL v2 weights are BSD-3 licensed [4]. These terms were read on 2026-10-28 and may change. An operator therefore faces a trade-off that our results make concrete: the model with the most consistent gains needs a commercial agreement, and the openly licensed one is significantly better than a tuned GBDT on only one of five datasets.

Table 7: Sensitivity (exploratory). (a) ΔEMP of each TFM against the best GBDT tuned with 25–40 trials and against the best of those and 200-trial LightGBM and CatBoost. (b) ΔEMP with duplicates removed (primary protocol) and with duplicates kept but split by duplicate group (five splits). Removed: copies discarded by de-duplication; Confl.: duplicate groups whose copies disagree on churn, with the rows they contain. p: unadjusted corrected t-test for the right-hand comparison.

(a) Tuning budget

Dataset	TFM	Ref. (25–40 trials)	ΔEMP	Ref. (incl. 200 trials)	ΔEMP	Seeds	p
Cell2Cell	TabPFN-3.5	LightGBM	+1.9%	LightGBM (200 trials)	+1.9%	5/5	0.005
Cell2Cell	TabICL v2	LightGBM	+0.5%	LightGBM (200 trials)	+0.4%	4/5	0.312
Iranian	TabPFN-3.5	LightGBM	+7.6%	LightGBM	+7.6%	10/10	<0.001
Iranian	TabICL v2	LightGBM	+7.4%	LightGBM	+7.4%	10/10	<0.001
Orange	TabPFN-3.5	LightGBM	+14.3%	LightGBM	+14.3%	10/10	0.011
Orange	TabICL v2	LightGBM	−4.5%	LightGBM	−4.5%	4/10	0.457

(b) Duplicates

Dataset	Removed	Confl. groups (rows)	Churn % (before/after)	TFM	Dedup.	Grouped	p
Telco	40	18 (42)	26.5 / 26.4	TabPFN-3.5	+1.0%	+0.8%	0.288
Telco	40	18 (42)	26.5 / 26.4	TabICL v2	+0.6%	+0.7%	0.322
Iranian	314	14 (64)	15.7 / 15.6	TabPFN-3.5	+7.6%	+7.5%	0.001
Iranian	314	14 (64)	15.7 / 15.6	TabICL v2	+7.4%	+6.9%	0.005
E-Commerce	556	0 (0)	16.8 / 16.6	TabPFN-3.5	+8.3%	+9.4%	0.035
E-Commerce	556	0 (0)	16.8 / 16.6	TabICL v2	+4.8%	+5.8%	0.148

Table 8: Protocols of the earlier benchmark [30] and of this replication on E-Commerce.

	Seyedzadeh & Karimi [30]	This work
Splits	5-fold stratified CV	60/20/20 holdout, 5 seeds
Duplicates	not stated	removed (556 rows)
GBDT baseline	XGBoost, library defaults	tuned LightGBM, XGBoost, CatBoost; default XGBoost
TFMs	TabICL v1 and v2, TabPFN v2, TabDPT, Mitra, LimiX, ConTextTab	TabICL v2, TabPFN-2.5, 3.5, 3.5-Fast
Metrics	F1, ROC-AUC, PR-AUC, log loss	PR-AUC, EMP, with corrected t-test

5 Discussion

The results support a narrower statement than "foundation models outperform trees for churn". With tuned baselines and a correction for multiple comparisons, one model generation, TabPFN-3.5, earns more campaign profit than a tuned GBDT on two of five datasets and is never behind; the other TFMs are rarely distinguishable from the GBDT. Three factors explain most of the variation. The model version matters more than the class. Baseline tuning removes part of any measured lead, about a third on E-Commerce. And the datasets most used in churn research, Telco and Maven, are those on which no model can be told apart, so studies built on them can neither confirm nor refute a TFM advantage.

For a practitioner this suggests an order of operations. Tune the GBDT first; default settings overstate what a TFM adds. Evaluate candidates on held-out campaign profit, not only AUC. If TabPFN-3.5 still leads and its licence terms are acceptable, score with a cached context. If only openly licensed models are possible, test TabICL v2 on one's own data and recalibrate it before thresholding. A balanced context is not a free improvement when the training set fits.

The finding most likely to change with more evidence is the size of the Orange gain: it is consistent across splits but not established after correction, and a feature-engineered GBDT [34] could narrow it. The finding least likely to change is the calibration pattern, which is consistent across splits and matches the underprediction reported for credit risk [10].

Table 9: Replication on E-Commerce (exploratory, five splits). PR-AUC mean (sd); Δ in PR-AUC points versus default XGBoost and versus the best tuned GBDT, and EMP relative to the best tuned GBDT. †: two-level bootstrap interval excludes zero.

Model	PR-AUC	EMP	ΔPR vs default	ΔPR vs tuned	ΔEMP
XGBoost (library default)	0.906 (0.019)	8.090	–	–	–
XGBoost	0.923 (0.019)	8.119	–	–	–
LightGBM	0.922 (0.016)	8.116	–	–	–
CatBoost	0.927 (0.017)	8.119	–	–	–
TabICL v2	0.967 (0.012)	8.505	**+6.1**†	**+4.0**†	**+4.8%**†
TabPFN-2.5	0.966 (0.012)	8.512	**+6.0**†	**+3.9**†	**+4.9%**†
TabPFN-3.5	0.984 (0.008)	8.790	**+7.8**†	**+5.8**†	**+8.3%**†

Table 10: Seconds to score 10k customers on one T4 (LightGBM on CPU), best of 10k and 50k batches. †: only the 1k batch was run because larger ones exceeded our time budget; per-customer time falls with batch size, so this is an upper bound. Context: training split (4,201 / 30,628 / 30,000 rows).

Model	Telco	Cell2Cell	Orange
LightGBM (1,000 trees, CPU)	0.8	0.5	0.8
TabICL v2	2.5	18.8	89.2
TabICL v2, KV cache	–	–	16.7
TabPFN-3.5	8.8	49.4	1,686†
TabPFN-3.5, cached context	8.1	30.5	44.9
TabPFN-3.5-Fast	2.7	12.4	25.9
TabPFN-2.5	9.6	3,065†	10,122†

6 Limitations

The evidence rests on three real-operator datasets. With 10 splits and a correction over 20 comparisons, the study has limited power: the Orange result is consistent in direction but not significant, and more splits or datasets could move it either way. We cannot claim that TabPFN-3.5's gains generalise to churn in other industries, or to operators with millions of customers, where the context would no longer hold the full training set.

The baselines were tuned with 25 to 40 trials; a fivefold budget changed nothing on three datasets, but we did no feature engineering, which can shift model comparisons [34], and we did not record the selected hyperparameters. Removing duplicate feature rows is a design choice; the grouped-split check covers it for the three affected datasets. The ablations, the sensitivity checks and the replication use five splits.

The cost model behind EMP and the policy uses the standard population-level parameters [36]; e-Profits gives a customer-level alternative on three datasets only. We used one GPU type and the default inference settings of each TFM, and did not evaluate the API-only TabPFN-3.5-Plus and -Thinking variants or the other TFMs covered by [30].

7 Conclusion

Measured with tuned baselines, in money, and with a correction for multiple comparisons, tabular foundation models pay off for churn selectively. TabPFN-3.5 earns significantly more campaign profit than a tuned GBDT on Cell2Cell and Iranian, wins in every split on Orange without surviving correction, and is never behind; its raw probabilities are also the best calibrated. Its weights, however, cannot be used in production without a commercial licence. TabICL v2 and TabPFN-2.5 are rarely distinguishable from a tuned GBDT, and TabICL v2 needs recalibration before its probabilities can drive a campaign. Profit metrics confirm clear

winners and reshuffle ties. We release the pipeline and every per-split prediction so that these conclusions can be re-checked as models and licences change.

Data and code availability

The version of the code and results described here is archived on Zenodo as release v1.3 (`https://doi.org/10.5281/zenodo.23738816`) and maintained at `https://github.com/ihpolash/tfm-churn-profit-audit`. The repository contains the prediction and analysis code as scripts and as notebooks, the Kaggle kernel definitions that ran them, the per-split predictions and split indices of all 685 runs, the analysis outputs, and the generator and audit for every table and number in this paper. The datasets are public (Table 2) but are not redistributed, because several are licensed for non-commercial or attributed use only. The labels needed to recompute metrics from the predictions are regenerated from the original sources by the released code.

Reproducibility

Every model run writes its predictions, split indices, timing and package versions; the analysis recomputes every metric from those files. Every table, figure and in-text number in this paper is generated by `paper/make_tables.py` from the released analysis outputs, and `paper/audit.py` re-derives them without rebuilding the pipeline. The audit checks that regeneration reproduces the committed tables exactly, recomputes every corrected t-test, Holm adjustment and intersection–union p-value from the per-split differences, confirms that each reference GBDT is the best on the calibration split, that all runs succeeded with identical and disjoint splits, that the robustness checks cover the datasets and splits stated, and that the package versions logged by every run match Appendix A. It needs only the Python standard library.

Declaration on the use of generative AI

The author used an AI assistant (Claude, Anthropic) for implementation and execution of the experiment code, for literature search and reference checking, for table and figure generation, and for drafting and editing this manuscript. The research question was chosen by the author; the benchmark design, evaluation protocol and interpretation of the results were developed in iterative discussion with the assistant, and every design decision and conclusion was reviewed and approved by the author. All numerical results were produced by the released code. Every table, figure and in-text number is generated by a released script from the released analysis outputs, which are computed from the released per-split predictions, and an audit notebook re-derives them before submission. Every reference was checked against its primary source or an authoritative index. The author takes full responsibility for all content of this paper, irrespective of how it was drafted.

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A Reproducibility Details

Table 11: Software, hardware and baseline search spaces. GBDTs: Optuna TPE, validation AUC on an inner 80/20 split, early stopping after 100 rounds (up to 3,000 trees), refit on the full training split.

Software	tabpfn 9.2.0 (TabPFN-2.5, 3.5, 3.5-Fast); tabicl 2.2.0 (checkpoint 2026-02-12); lightgbm 4.6.0; xgboost 3.4.1; catboost 1.2.10; interpret 0.7.8; optuna 5.0.0; scikit-learn 1.6.1; torch 2.11.0 (CUDA 12.8)
Hardware	Kaggle: NVIDIA T4 (16 GB) for TFMs; CPU sessions for all other models
LightGBM	40 trials (200 in Section 4.5); learning rate $[0.005, 0.1]$ log; leaves $[8, 128]$ log; min child samples $[5, 200]$ log; subsample $[0.5, 1]$; column sample $[0.3, 1]$; L2 $[10^{-3}, 10]$ log
XGBoost	40 trials; learning rate $[0.005, 0.1]$ log; depth $[3, 10]$; min child weight $[0.5, 50]$ log; subsample $[0.5, 1]$; column sample $[0.3, 1]$; L2 $[10^{-3}, 10]$ log
CatBoost	25 trials (200 in Section 4.5); learning rate $[0.01, 0.2]$ log; depth $[4, 10]$; L2 $[1, 30]$ log; random strength $[0.1, 10]$ log; bagging temperature $[0, 1]$
EBM, LR	defaults; LR with one-hot encoding (up to 30 categories), median imputation with missing indicators, scaling

B Additional Results

Table 12: Exploratory: PR-AUC difference in points (×100) and EMP difference relative to the best tuned GBDT, over 10 splits. $*$: significant after Holm correction (EMP, primary family only); †: two-level bootstrap interval excludes zero. Reference shown as PR-AUC / EMP (LGBM: LightGBM, XGB: XGBoost, CatB: CatBoost).

	Telco		Maven		Cell2Cell		Iranian		Orange	
	vs. XGB / LGBM		vs. CatB / XGB		vs. XGB / LGBM		vs. LGBM / LGBM		vs. CatB / LGBM	
TFM	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP	ΔPR	ΔEMP
TabICL v2	**+0.6**†	**+0.6**%†	+0.3	+0.3%	**+0.5**†	**+0.5**%†	**+4.6**†	**+7.4**%*	0.0	−4.5%
TabPFN-2.5	+0.5	**+0.6**%†	+0.2	**+0.7**%†	**+0.6**†	**+0.4**%†	**+4.4**†	**+6.9**%*	−0.5	−3.0%
TabPFN-3.5	**+0.7**†	**+1.0**%†	**+1.2**†	**+1.0**%†	**+2.9**†	**+1.9**%*	**+5.0**†	**+7.6**%*	**+2.8**†	**+14.3**%†
TabPFN-3.5-Fast	**+0.7**†	**+0.9**%†	**+0.9**†	+0.7%	**+2.0**†	**+1.1**%*	**+4.9**†	**+7.1**%*	**+1.9**†	**+7.8**%†

Table 13: Mean PR-AUC and EMP (profit per customer, CLV units) over 10 splits; best per column in bold.

	Telco		Maven		Cell2Cell		Iranian		Orange	
Model	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP	PR-AUC	EMP
Logistic regression	0.647	10.294	0.830	12.769	0.388	8.654	0.754	6.356	0.173	0.409
EBM	0.661	10.246	0.894	13.076	0.460	9.117	0.936	7.699	0.194	0.534
LightGBM	0.657	10.232	0.890	13.067	0.468	9.176	0.941	7.860	0.206	0.576
XGBoost	0.659	10.241	0.891	13.059	0.466	9.175	0.936	7.791	0.200	0.565
CatBoost	0.659	10.215	0.893	13.070	0.466	9.151	0.935	7.709	0.210	0.557
TabICL v2	0.666	10.298	0.895	13.101	0.472	9.218	0.987	8.444	0.210	0.550
TabPFN-2.5	0.664	10.292	0.894	13.149	0.473	9.208	0.985	8.403	0.204	0.559
TabPFN-3.5	**0.667**	**10.336**	**0.904**	**13.194**	**0.495**	**9.350**	**0.990**	**8.457**	**0.238**	**0.658**
TabPFN-3.5-Fast	0.666	10.322	0.902	13.154	0.487	9.280	0.989	8.421	0.229	0.621

Table 14: Mean ROC-AUC over 10 splits; best per column in bold.

Model	Telco	Maven	Cell2Cell	Iranian	Orange
Logistic regression	0.845	0.920	0.620	0.932	0.702
EBM	0.846	0.944	0.682	0.983	0.728
LightGBM	0.844	0.942	0.685	0.987	0.734
XGBoost	0.846	0.942	0.685	0.985	0.733
CatBoost	0.845	0.943	0.684	0.985	0.728
TabICL v2	0.849	0.945	0.687	0.997	0.727
TabPFN-2.5	0.848	0.946	0.688	0.996	0.733
TabPFN-3.5	**0.850**	**0.950**	**0.702**	**0.997**	**0.748**
TabPFN-3.5-Fast	0.849	0.948	0.697	0.997	0.742